Evaluation

C4VHMI

SECTION – A

Choose the correct answer

1. A loop invariant need not be true

   (a) at the start of the loop. (b) at the start of each iteration

   (c) at the end of each iteration (d) at the start of the algorithm

2. We wish to cover a chessboard with dominoes, ▭ the number of black squares and the number of white squares covered by dominoes, respectively, placing a domino can be modeled by

   (a) b := b + 2 (b) w := w + 2 (c) b, w := b+1, w+1 (d) b := w

3. If m x a + n x b is an invariant for the assignment a, b : = a + 8, b + 7, the values of m and n are

   (a) m = 8, n = 7 (b) m = 7, n = -8 (c) m = 7, n = 8 (d) m = 8, n = -7

4. Which of the following is not an invariant of the assignment?

   m, n := m+2, n+3

   (a) m mod 2 (b) n mod 3 (c) 3 X m - 2 X n (d) 2 X m - 3 X n

5. If Fibonacci number is defined recursively as

$$\mathbf{F(n)}= \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ F(n-1)+F(n-2) & \text{otherwise} \end{cases}$$

   to evaluate F(4), how many times F() is applied?

   (a) 3 (b) 4 (c) 8 (d) 9

6. Using this recursive definition

$$\mathbf{a}^{\mathbf{n}}= \begin{cases} 1 & \text{if } n = 0 \\ a \times a^{n-1} & \text{otherwise} \end{cases}$$

   how many multiplications are needed to calculate a^{10}?

   (a) 11 (b) 10 (c) 9 d) 8

SECTION-B

Very Short Answers

1. What is an invariant?
2. Define a loop invariant.
3. Does testing the loop condition affect the loop invariant? Why?
4. What is the relationship between loop invariant, loop condition and the input- output recursively

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